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Brief paper

Finite-time trajectory tracking control in a task space of robotic manipulators☆

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ABSTRACT

This work addresses the problem of the accurate task space control subject to finite-time convergence. Dynamic equations of a rigid robotic manipulator are assumed to be uncertain. Moreover, globally unbounded disturbances are allowed to act on the manipulator when tracking the trajectory by the end-effector. Based on suitably defined task space non-singular terminal sliding vector variable and the Lyapunov stability theory, we derive a class of absolutely continuous Jacobian transpose robust controllers, which seem to be effective in counteracting uncertain dynamics, unbounded disturbances and (possible) kinematic and/or algorithmic singularities met on the end-effector trajectory. The numerical simulations carried out for a robotic manipulator of a SCARA type consisting of two revolute kinematic pairs and operating in a two-dimensional task space, illustrate performance of the proposed controllers.
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1. Introduction

In recent years, interest has increased in applying robotic manipulators to useful practical tasks requiring extremely high precision and stability of the performance. In most situations met in practice, such tasks are specified in terms of a trajectory expressed in Cartesian coordinates to be tracked by the end-effector. In order to apply known joint space control techniques (see e.g. our recent work Galicki, 2015) for tracking such a trajectory, an inverse or pseudo-inverse kinematics algorithm has to be utilized. The process of kinematic inversion is both time consuming (there does not exist, in general, an analytic form of inverse mapping) and becomes very complicated when the Cartesian trajectory generates kinematic and/or algorithmic singularities (Balleieul, 1985). Thus, a controller to be designed should accurately track desired end-effector trajectory despite possible singularities met on this trajectory, uncertain dynamic equations, unknown payload to be transferred by the end-effector and external disturbances. Moreover, such controller has to generate at least absolutely continuous control signals (torques) to avoid undesirable chattering. Due to the challenging nature of the aforementioned control design problems, many researchers have proposed different types of controllers. In such a context, one can distinguish three major approaches of controlling the robotic manipulators in the task space. The control techniques offered in the first approach (Kelly & Moreno, 2005; Moreno-Valenzuela & Gonzales-Hernandez, 2011; Nakanishi, Cory, Mistry, Peters, & Schaal, 2008; Ott, Dietrich, & Schaffer, 2015; Siciliano, Sciavicco, Villani, & Oriolo, 2010) require the full knowledge of the dynamics neglecting the external disturbances. In the second approach, works (Braganza, Dixon, Dawson, & Xian, 2008; Cheah, Liu, & Slotine, 2006; Colbaugh & Glass, 1995; Galicki, 2014; Li & Cheah, 2013; Tatlicioglu, Braganza, Burg, & Dawson, 2008) propose adaptive control algorithms to compensate for parametric uncertainties in dynamic model including only the linearly parametrizable friction terms (viscous friction) and also neglecting the external (non-linearly parametrizable) disturbances (except of Colbaugh & Glass, 1995 which permits globally bounded disturbances). In the third approach, model based robust control scheme was proposed in work (Ozbay, Sahin, & Zergeroglu, 2008). In addition, all the aforementioned control schemes (except of Galicki, 2014) require explicit inverse or pseudo-inverse of a Jacobian matrix, which may result in numerical instabilities due to (possible) kinematic and/or algorithmic singularities (Balleieul, 1985). Furthermore, all the control schemes assume globally bounded disturbances when tracking the trajectory whereas e.g. a viscous friction term is globally unbounded. Finally, all the aforementioned controllers provide only at most asymptotic stability what may be insufficient for accomplishment of tasks requiring the extremely high precision (e.g. assembly of electronic components on the small surface of printed circuit boards). Consequently, all those algorithms are not able to generate continuous controls resulting

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in finite-time stability of the equilibrium when (possible) singular configurations may appear on the trajectory, dynamic equations are uncertain and (unbounded) disturbances act on the robotic manipulator. In this study, a new task space non-singular terminal sliding manifold (TSM) is introduced to track the end-effector trajectory. The proposed TSM manifold makes it possible to simultaneously join the first order sliding mode approach possessing the finite-time control capabilities with the second order sliding mode techniques generating the (absolutely) continuous controls. The solution of the tracking control problem is based herein on introducing a new dynamic version of a static computed torque approach presented e.g. in works (Siciliano et al., 2010; Spong & Vidyasagar, 1989). By fulfilment of reasonable assumption regarding the Jacobian matrix, the proposed Jacobian transpose control scheme is shown to be finite-time stable. The remainder of the paper is organized as follows. Section 2 formulates the finite-time trajectory tracking task. Section 3 sets up a class of task space robust absolutely continuous controllers solving the trajectory tracking problem in a finite-time subject to uncertain robot dynamic equations and unbounded disturbances. Section 4 presents computer examples of the end-effector trajectory tracking by a robotic manipulator of a SCARA type, consisting of two revolute kinematic pairs and operating in two-dimensional task space. Finally, some concluding remarks are drawn in Section 5. Throughout this paper, $\lambda_{\max}(\cdot), \lambda_{\min}(\cdot)$ denote the maximal and minimal, respectively, eigenvalues of the symmetric matrices $(\cdot)$.

2. Problem formulation

The robust control scheme designed in the next section is applicable to holonomic mechanical systems comprising robotic manipulators considered here which are described, in general, by the following dynamic equations, expressed in generalized (joint) coordinates $q \in \mathbb{R}^n$ (Spong & Vidyasagar, 1989):

$$M(q)\ddot{q} + H(q,\ \dot{q}) + G(q) + D(t,\ q,\ \dot{q}) = v, \tag{1}$$

where $\dot{q}$ and $\ddot{q}$ represent the velocity and acceleration, respectively. The $n \times n$ inertia matrix $M(q)$ is positive definite and symmetric. $H = B(q)(\dot{q}\cdot\dot{q}) + C(q)(\dot{q}^2)$, where B and C are the $n \times \frac{n(n-1)}{2}$ and $n \times n$ matrices of coefficients of the Coriolis and centrifugal forces, respectively. $(\dot{q}\cdot\dot{q}) = (\dot{q}_1\dot{q}_2, \ldots, \dot{q}_{n-1}\dot{q}_n)^T$ and $(\dot{q}^2) = (\dot{q}_1^2, \ldots, \dot{q}_n^2)^T$, respectively. $v = (v_1, \ldots, v_n)^T$ stands for the n-dimensional vector of controls (torques/forces). $G(q)$ is the n-dimensional vector of generalized gravity forces. $D(t,\ q,\ \dot{q})$ means the n-dimensional external disturbance signal which is (by assumption) at least absolutely continuous with $\dot{D}(t,\ q,\ \dot{q})$ as being a locally bounded Lebesgue measurable mapping. Moreover, $\|D\|$ and $\|\dot{D}\|$ are (by assumption) upper estimated as follows $\|D\| \le \alpha_0(t),\ \|\dot{D}\| \le \alpha_1(t)$, where α_0, α_1 stand for the known, non-negative functions. The general kinematic and differential mappings between joint coordinates q and task ones $p \in \mathbb{R}^m$ can be written as

$$p = f(q), \qquad \dot{p} = J\dot{q}, \tag{2}$$

where $n \ge m$ is the dimension of the Cartesian space in which the end-effector operates; $f : \mathbb{R}^n \longrightarrow \mathbb{R}^m$ and $J = \frac{\partial f}{\partial q}$ is the $m \times n$ Jacobian matrix. Due to the fact that kinematic redundancy is not significant in the design of our controller, m is assumed to be equal to n. A task accomplished by the end-effector consists in tracking a desired trajectory $p_d(t) \in \mathbb{R}^n$, $t \in [0,\ \infty)$ which is assumed to be at least triply continuously differentiable, i.e., $p_d(\cdot) \in C^3[0,\ \infty)$. By introducing the task tracking error $e = f(q) - p_d$, we may formally express the finite-time control problem by means of the following equations:

$$\lim_{t\to T} e(t) = \lim_{t\to T} \dot{e}(t) = \lim_{t\to T} \ddot{e}(t) = 0, \tag{3}$$

where $0 \le T$ denotes a finite-time of convergence of $f(q)$ to p_d. In further analysis, J is assumed to be of the full rank in the operation region, i.e.,

$$\operatorname{rank}(J(q)) = n. \tag{4}$$

Let us note that condition (4) may be made somewhat more weak. It suffices that for $0 \ne x \in \mathbb{R}^n$ and singular configuration q', the following condition holds true: $x \notin \ker(J^T(q'))$. In the sequel, useful properties of (1) are summarized which will be utilized while designing the controller. The following inequalities are satisfied (Spong & Vidyasagar, 1989):

$$0 < \|M^{-1}\|_F \le \Lambda_{\max}, \qquad \|B\|_F + \|C\|_F \le c_1, \qquad \|G\| \le c_2, \tag{5}$$

where $\|\ \|_F$ means the Frobenius (Euclidean) matrix norm; c_1, c_2, $\Lambda_{\max}$ are known positive scalar coefficients. Moreover, the following inequalities hold true for revolute kinematic pairs:

$$\|J(q)\|_F \le c_3, \qquad \left\|\frac{\partial J}{\partial q}\right\|_F \le c_3', \qquad \left\|\frac{\partial^2 J}{\partial q^2}\right\|_F \le c_3'', \tag{6}$$

where c_3, c_3', c_3'' are known scalar coefficients. Furthermore, from (4) and (5), one also obtains that

$$0 < \lambda\mathbb{I}_n \le JM^{-1}J^T \le \Lambda\mathbb{I}_n, \tag{7}$$

where λ, Λ denote estimations of minimal and maximal, respectively eigenvalues of matrix $JM^{-1}J^T$; $\mathbb{I}_n$ stands for the $n \times n$ identity matrix. In order to obtain at least absolutely continuous control v, let us differentiate (1) with respect to time thus obtaining $M(q)\frac{d^3q}{dt^3} + F(q,\ \dot{q},\ \ddot{q},\ t) = \dot{v}$, where $F = \dot{M}\ddot{q} + \dot{B}(\dot{q}\cdot\dot{q}) + \dot{C}(\dot{q}^2) + B\frac{d}{dt}(\dot{q}\cdot\dot{q}) + C\frac{d}{dt}(\dot{q}^2) + \dot{G} + \dot{D}$. Motivated in part by the static computed torque methodology (Siciliano et al., 2010; Spong & Vidyasagar, 1989), we propose a new dynamically computed torque vector $\dot{v}$ of the form

$$\dot{v} = J^T\hat{M}(q)u + \hat{F}(q,\ \dot{q},\ \ddot{q},\ t), \tag{8}$$

where $\hat{M}$ and $\hat{F}$ denote known estimates of the corresponding unknown terms M and F, respectively; $u \in \mathbb{R}^n$ is a new control to be found. If F is known mapping, we can take $\hat{F} = F$. Alternatively, $\hat{F} = 0$ if no model of F is available. Definition of $\hat{M}$ is given in the next section. In the sequel, we introduce the following auxiliary matrix $\mathcal{R} = JM^{-1}J^T\hat{M}$ which will play a crucial role by designing our controller. Let us triply differentiate e with respect to time thus obtaining $\frac{d^3e}{dt^3} = u + (\mathcal{R} - \mathbb{I}_n)u + Q - \frac{d^3p_d}{dt^3}$, where $Q = JM^{-1}(\hat{F} - F) + \ddot{J}\dot{q} + 2\dot{J}\ddot{q}$. Furthermore, based on definition of Q, an upper estimation on $\|Q\|$ takes the form

$$\|Q\| \le \mathcal{W}(t,\ q,\ \dot{q},\ \ddot{q}), \tag{9}$$

where $\mathcal{W} = c_3\Lambda_{\max}\|\hat{F}\| + c_4\|\dot{q}\|\|\ddot{q}\| + c_5\|\dot{q}\|^3 + c_6\|\dot{q}\| + c_3\Lambda_{\max}\alpha_1$; c_4, c_5 and c_6 are (known by assumption) positive scalar coefficients for which the following inequalities hold true: $c_4 \ge \|JM^{-1}\|_F(\|\frac{\partial M}{\partial q}\|_F + \|B\|_F + \|C\|_F) + 3\|\frac{\partial J}{\partial q}\|_F$; $c_5 \ge \|JM^{-1}\|_F(\|\frac{\partial B}{\partial q}\|_F + \|\frac{\partial C}{\partial q}\|_F) + \|\frac{\partial^2 J}{\partial q^2}\|_F$ and $c_6 \ge \|JM^{-1}\|_F\|\frac{\partial G}{\partial q}\|_F$. Based on (8), the next section will present an approach to the solution of the control problem (1), (3) making use of the Lyapunov stability theory.

3. Control of the robotic manipulator

In the sequel, we start the analysis of a controller design by the assumption that joint positions, velocities and accelerations are available from measurements. Based on (7), we can make the following remark:

$$(\exists\hat{M} > 0)(\exists\rho > 0)(|\lambda_{\max}(\mathcal{R} - \mathbb{I}_n)| \le \rho < 1). \tag{10}$$

If we set $\hat{M} = \frac{2}{\Lambda+\lambda}\mathbb{I}_n$ then $\rho = \frac{\Lambda-\lambda}{\Lambda+\lambda}$ satisfies inequality (10). Let $s \in \mathbb{R}^n$ be the following non-singular terminal sliding manifold expressed in task coordinates:

$$s = \ddot{e} + \int_0^t \left(\lambda_2 \ddot{e}^{3/5} + \lambda_2 \lambda_1^{3/5} (\dot{e}^{9/7} + \lambda_0^{9/7} e)^{1/3}\right) d\tau, \tag{11}$$

where $\lambda_0, \lambda_1, \lambda_2$ stand for positive definite diagonal gain matrices. The potency of both $e, \dot{e}, \ddot{e}$ and $\lambda_0, \lambda_1, \lambda_2$ is defined component-wise. In order to fulfil equality constraints (3) subject to (8), a (simple) robust task space control law is proposed as follows

$$\dot{v} = J^T \hat{M}(u_n + u_r) + \hat{F}(q,\ \dot{q},\ \ddot{q},\ t), \tag{12}$$

where $u_n = \frac{d^3 p_d}{dt^3} - \lambda_2 \ddot{e}^{3/5} - \lambda_2 \lambda_1^{3/5} (\dot{e}^{9/7} + \lambda_0^{9/7} e)^{1/3} - \Lambda_n s^\beta$; $u_r = \begin{cases} -\frac{\kappa}{1-\rho}\frac{s}{\|s\|}(\rho\|u_n\| + \mathcal{W}) & \text{for } s \neq 0 \\ 0 & \text{otherwise,} \end{cases}$; Λ_n denotes positive definite gain matrix; $\beta = \frac{a}{b}$; a, b are positive odd numbers, $a < b < 2a$. The solution of (12) is meant in the Filippov sense (Filippov, 1988). Applying the Lyapunov stability theory, we now derive the following result.

Theorem 1. *If Jacobian matrix J fulfils (4) in a (closed) domain of our interest, q, $\dot{q}$, $\ddot{q}$ are directly available and $\lambda_0, \lambda_1, \lambda_2, \Lambda_n > 0$, $\kappa > 1$ then control scheme (12) guarantees stable convergence in a finite time of the task errors $(e,\ \dot{e},\ \ddot{e})$ to the origin $(e,\ \dot{e},\ \ddot{e}) = (0,\ 0,\ 0)$.*

Proof. Since the proof of this theorem is similar to that given in our recent work (Galicki, 2015), we limit ourselves herein to the key steps of argumentation. Consider Lyapunov function candidate $V = \frac{1}{2}\langle s,\ s\rangle$ whose time derivative after simple algebra equals $\dot{V} = -\langle s,\ \Lambda_n s^\beta\rangle + \langle s, u_r\rangle + \langle s, (\mathcal{R} - \mathbb{I}_n)(u_n + u_r) + Q\rangle$, where $\langle\ ,\ \rangle$ stands for the scalar product of vectors. Taking into account the assumption of the full rank Jacobian matrix and inequalities (7), we can easily deduce that $\dot{V} \leq -\lambda_{\min}(\Lambda_n) 2^{\frac{1+\beta}{2}} V^{\frac{1+\beta}{2}}$. Finally, we conclude that the origin $(e,\ \dot{e},\ \ddot{e}) = (0,\ 0,\ 0)$ is attainable in a finite time T. □

A few remarks may be made regarding the control law (12) and Theorem 1.

- Remark 1. Let us note that expression (12) presents also a transposed Jacobian controller. In such a context, the use of the transpose of the Jacobian is a well-known technique and there are several papers (Cheah, 2006; Cheah, Lee, Kawamura, & Arimoto, 2000; Kelly, 1996; Moosavian & Papadopoulos, 2007; Siciliano, 1990; Wolovich & Elliot, 1984) that include its stability analysis. Nevertheless, work (Wolovich & Elliot, 1984) guarantees only ultimate boundedness provided that $\dot{p}_d(t)$ is globally norm bounded. The asymptotically stable purely kinematic control scheme offered in Siciliano (1990) indeed eliminates explicit computation of the inverse but introduces undesirable chattering effect. In work (Cheah, 2006), it is claimed that controllers based on transposed Jacobian and inverse Jacobian are dual in the sense that the transformation from task space to joint space can be either defined as transposed Jacobian or inverse Jacobian. As was also shown in Cheah (2006) and Cheah et al. (2000), approximate transpose Jacobian control law is asymptotically stable. However, works (Cheah, 2006; Cheah et al., 2000; Kelly, 1996; Moosavian & Papadopoulos, 2007) provide stability analysis for the set-point control problems. On the other hand, Theorem 1 provides stability analysis for the trajectory tracking problems. Moreover, chattering-free transposed Jacobian controller (12) is able to attain the stable equilibrium $(e,\ \dot{e},\ \ddot{e}) = (0,\ 0,\ 0)$ in a finite time. Due to involving the sliding mode term u_r, controller (12) is also robust against uncertainties of dynamic equations and external unbounded disturbances.
- Remark 2. Observe that term $\langle s,\ \mathcal{R}u_r\rangle$ in $\dot{V}$ can be transformed after simple algebra as follows $\langle s,\ \mathcal{R}u_r\rangle = -\kappa\langle J^T s,\ M^{-1}J^T s\rangle \frac{2}{\Lambda+\lambda}(\rho\|u_n\| + \mathcal{W})$. Let J be singular at manipulator configuration q' and $0 \neq s \notin \ker(J^T(q'))$. Hence, for sufficiently large κ, term $\langle s,\ \mathcal{R}u_r\rangle$ can take arbitrarily large negative values which implies negative value of $\dot{V}$. Consequently, controller (12) is able to generate manipulator motions which can pass through singular manifold $\{q' :\ \det(J(q')) = 0\}$ as the computer simulations carried out in the next section show.

In most cases, real robotic manipulators are equipped with encoders which measure only joint positions. Hence, reconstruction or estimation of joint velocity and acceleration is required to apply controller (12). High-frequency measurement noise related with application of backward difference estimation or additional phase lag introduced by the filtering techniques (Berghuis & Nijmeijer, 1993) may be harmful to the closed-loop system (1), (12). Recently, a method of direct lag-free joint acceleration sensing has been proposed in work (Vihonen, Honkakorpi, Mattila, & Visa, 2013). However, its combination with our controller requires further investigations. Application of Luenberger-style observers (Canudas de Wit, Fixot, & Astrom, 1992; ElBeheiry, Zaki, & ElMaraghy, 2003), high-gain observers (Ball & Khalil, 2013; Khalil & Praly, 2014), model-free observers (De Luca, Schroder, & Thummel, 2007; Hsiao & Weng, 2013) or a class of observers based on the sliding-mode algorithms (Davila, Fridman, & Levant, 2005) seems to be an efficient approach compared with both backward difference technique and direct measurement. Although all the aforementioned observers are able to reliably reconstruct manipulator state (both joint velocity and acceleration) based on position measurement, there appears a difficulty to combine our control law and an observer from Ball and Khalil (2013), Canudas de Wit et al. (1992), De Luca et al. (2007), ElBeheiry et al. (2003), Hsiao and Weng (2013) and Khalil and Praly (2014). In order to make such combination possible, observers proposed in works (Ball & Khalil, 2013; Canudas de Wit et al., 1992; De Luca et al., 2007; ElBeheiry et al., 2003; Hsiao & Weng, 2013; Khalil & Praly, 2014) have to satisfy the so-called separation principle (Atasi & Khalil, 2000) which implies both the continuity of the controllers from Ball and Khalil (2013), Canudas de Wit et al. (1992), De Luca et al. (2007), ElBeheiry et al. (2003), Hsiao and Weng (2013) and Khalil and Praly (2014) with the fully available state and asymptotic stability of the closed-loop system under the continuous state feed-back controllers. Let us observe that our control law (12) is discontinuous what prevents an application of the state observers from Ball and Khalil (2013), Canudas de Wit et al. (1992), De Luca et al. (2007), ElBeheiry et al. (2003), Hsiao and Weng (2013) and Khalil and Praly (2014). Although the observer offered in Davila et al. (2005) fulfils the separation principle, our controller handles unbounded uncertainties (in dynamics and disturbances) and does not require boundedness of $\dot{q}$ and $\ddot{q}$, respectively. A computationally efficient approach based on the uniform robust exact finite-time differentiation has been recently proposed in work (Levant & Livne, 2012) to numerically find derivatives of absolutely continuous functions. The separation principle is trivially fulfilled for differentiators (model-free observers) from Levant and Livne (2012). Assuming that position $q = q(t)$ is known (measurable), one can exactly reconstruct both velocity $\dot{q}(t)$ and acceleration $\ddot{q}(t)$ (by neglecting the measurement noise of a device) after a finite-time transient process, say $T' > 0$. The second-order uniform robust exact differentiator takes in our case the following form:

$$\begin{aligned} \dot{y}_0 &= y_1 - \hat{\lambda}_2 L(t)^{1/3}|y_0 - q|^{2/3}\text{sign}(y_0 - q), \\ \dot{y}_1 &= y_2 - \hat{\lambda}_1 L(t)^{2/3}|y_0 - q|^{1/3}\text{sign}(y_0 - q), \\ \dot{y}_2 &= -\hat{\lambda}_0 L(t)\text{sign}(y_0 - q), \end{aligned} \tag{13}$$

where $\hat{\lambda}_0, \hat{\lambda}_1, \hat{\lambda}_2$ are positive constants. y_1, y_2 denote the outputs of differentiator (13) exactly reconstructing velocity $\dot{q}(t)$ and acceleration $\ddot{q}(t)$, i.e., $\dot{q}(t) = y_1(t)$, $\ddot{q}(t) = y_2(t)$ for $t \geq T'$. $L(t)$ stands for a positive continuous function which takes the form $L(t) = L'(t) + L''(t)$, where $L' = \Lambda_{\max}[c_3\Lambda(\|u_n\| + \frac{\kappa}{1-\rho}(\rho\|u_n\| + \mathcal{W})) + \|\hat{F}\|]$; $L'' = \Lambda_{\max}\{\|y_1\|(\|\frac{\partial M}{\partial q}\|_F + \|B\|_F + \|C\|_F)\Lambda_{\max}[\|v\| + c_1\|y_1\|^2 + c_2 + \alpha_0(t)] + (\|\frac{\partial B}{\partial q}\|_F + \|\frac{\partial C}{\partial q}\|_F)\|y_1\|^3 + \|\frac{\partial G}{\partial q}\|_F\|y_1\| + \alpha_1(t)\}$. $L(t)$ represents physically an upper estimation of the norm of $\frac{d^3q}{dt^3}$ (manipulator jerk). Let us define concatenating control v_c as follows

$$v_c = \begin{cases} v'(t), & t \in [0,\ T'], \\ v(t) & \text{given by (12)}, \dot{q} = y_1\ \ddot{q} = y_2, t > T', \end{cases} \tag{14}$$

where $v'(t)$ is arbitrary absolutely continuous mapping of time t (e.g. $v'(t) = 0$). Based on (13) and (14), we are now in a position to give the following theorem.

Theorem 2. *If Jacobian matrix J fulfils (4) in a (closed) operation region of our interest, q is only available from measurements and λ_0, λ_1, λ_2, $\Lambda_n > 0$, $\kappa > 1$ then control scheme (14) guarantees stable convergence in a finite time of the task errors $(e,\ \dot{e},\ \ddot{e})$ to the origin $(e,\ \dot{e},\ \ddot{e}) = (0,\ 0,\ 0)$.*

Proof. The proof of this theorem is similar to that given in our work (Galicki, 2015). Therefore, it is omitted herein. □

In a general case, if measured position $q = q(t)$, obtained from encoder, is additionally contaminated by a measurement noise $\eta(t)$, i.e., $q(t) = q_0(t) + \eta(t)$, where $\|\eta\| \leq \epsilon L(t)$; ϵ denotes a normalized noise magnitude (practically $\epsilon \in [10^{-6},\ 10^{-5}]$); $q_0(t)$, $\dot{q}_0(t)$, $\ddot{q}_0(t)$ stand for unknown true (noise-free) joint position, velocity and acceleration then observer (13) should also be applied to estimate quantities $e, \dot{e}$ and $\ddot{e}$. Notice from Eqs. (13) that $\|q_0(t) - y_0(t)\| \leq L(t)O(\epsilon)$, $\|\dot{q}_0(t) - y_1(t)\| \leq L(t)O(\epsilon^{2/3})$, $\|\ddot{q}_0(t) - y_2(t)\| \leq L(t)O(\epsilon^{1/3})$ after a finite transient time (Levant & Livne, 2012). Moreover, we know from Theorem 1 that $\|f(q_0) - p_d(t)\| \to 0$, $\|\dot{f}(q_0) - \dot{p}_d(t)\| \to 0$, $\|\ddot{f}(q_0) - \ddot{p}_d(t)\| \to 0$ after a finite transient time, too. Hence, for estimations $\hat{e} = f(y_0) - p_d$, $\dot{\hat{e}} = J(y_0)y_1 - \dot{p}_d$ and $\ddot{\hat{e}} = J(y_0)y_2 + \frac{dJ(q)}{dt}\Big|_{\substack{q=y_0\\ \dot{q}=y_1}} y_1 - \ddot{p}_d$, we obtain after simple algebra that $\|\hat{e}\| \leq c_3L(t)O(\epsilon)$; $\|\dot{\hat{e}}\| \leq c_3'\|y_1\|L(t)O(\epsilon) + c_3L(t)O(\epsilon^{2/3})$; $\|\ddot{\hat{e}}\| \leq (c_3'\|y_2\| + c_3''\|y_1\|^2)L(t)O(\epsilon) + c_3'(2\|y_1\|)L(t)O(\epsilon^{2/3}) + c_3L(t)O(\epsilon^{1/3})$.

4. Numerical examples

This section demonstrates the performance of the controller given by expressions (13), (14) on the three selected manipulator tasks. For this purpose, we utilize the data of the EDDA manipulator ($n = 2$) whose components of dynamic equations are given in Galicki (2015). Dynamic parameters $\Theta \in \mathbb{R}^4$ take the following nominal values: $\Theta_n = (3.1\ \ 9.5\ \ 0.24\ \ 0.77)^T$. The link lengths l_1, l_2 are equal to $l_1 = 0.3$, $l_2 = 0.58$, respectively. The initial configuration and velocity of the manipulator are equal in all the experiments to $q(0) = (-\pi/4,\ \pi/4)^T$, $\dot{q}(0) = (0,\ 0)^T$, respectively. Due to the planar manipulator of the two revolute kinematic pairs taken for simulations, there exists analytic solution to the inverse kinematics problem, i.e., $q = f^{-1}(p_d(t))$, $t \in [0,\ \infty)$, where f^{-1} is given analytically. Hence, based on nominal parameters Θ_n, we numerically find the following quantities: $\Lambda' = \sup_{t\geq 0}\{\lambda_{\max}(J(q)M^{-1}(q)J^T(q))|_{q=f^{-1}(p_d(t))}\}$, $\lambda' = \inf_{t\geq 0}\{\lambda_{\min}(J(q)M^{-1}(q)J^T(q))|_{q=f^{-1}(p_d(t))}\}$ which could directly serve as estimates for Λ and λ, respectively, in the case of non-singular p_d (i.e. for non-singular configurations only). Observe that for singular configuration q', we have $\lambda' = 0$ which would prevent fulfilment of inequalities (7) although controller (14) may work in such a case (see Remark 2. and the second numerical experiment). In order to apply controller (14) for both non-singular and singular desired end-effector trajectories p_d, quantities Λ', λ' corresponding to non-singular end-effector trajectory were slightly varied (Λ' was increased and λ' was decreased by a small number (found experimentally), respectively). Consequently, our estimates for controller (14) are chosen as $\hat{M} = \frac{2}{\lambda+\Lambda}\mathbb{I}_2$; $\hat{F} = 0$; $\lambda = 0.17$; $\Lambda = \Lambda_{\max} = 5$. In order to speed up the convergence process, we have chosen good initial guesses $y_1(0), y_2(0)$ in the numerical examples (which imply relation $T' \simeq 0$) based on the nominal values of our dynamic model. Consequently, differentiator (13) was run with the following initial values: $y_1(0) = \dot{q}(0)$, $y_2(0) = \ddot{q}(0) = (-22.1,\ 5.7)^T$, $v(0) = (0,\ 0)^T$ and parameters $\hat{\lambda}_0 = 13.2$, $\hat{\lambda}_1 = 16.8$, $\hat{\lambda}_2 = 24$. In order to simplify numerical computations, rough conservative estimations of c_i, $i = 1:6$ have been assumed as follows $c_1 = 10$, $c_2 = 150$, $c_3 = 1$, $c_4 = 27$, $c_5 = 5$ and $c_6 = 10$, respectively. To show the robustness of our control scheme against singular configurations, the manipulator is to accomplish three tasks. The aim of the controller (14) in the first task is to track by the end-effector a circle trajectory, expressed by the following equation: $p_d^1(t) = (0.5\cos(t),\ 0.1 + 0.5\sin(t))^T$. Disturbing term D is assumed in this simulation to represent both the friction in the sliding (viscous) and in the presliding (also called stiction) regime (Canudas de Wit, Ollson, Astrom, & Lischinsky, 1995) of the form $D = \theta_5(q - q(0)) + \theta_6\dot{q}$ with the nominal values of parameters θ_5, θ_6 equal to $\theta_{5,n} = \theta_{6,n} = 2$. Moreover, $\dot{D}$ is equal to $\dot{D} = \theta_5\dot{q} + \theta_6\ddot{q}$. The control law proposed in Ozbay et al. (2008) for linearly parametrizable continuous term D is given by the following expression:

$$v_c = Y\hat{\Theta} + Kr - J^Te + \hat{\theta}_5(q - q(0)) + \hat{\theta}_6\dot{q}, \tag{15}$$

where $r = \dot{q} - J^{-1}(\dot{p}_d - \alpha e)$ denotes a filtered tracking error signal (Ozbay et al., 2008); $Y = Y(e, \dot{e}, q, \dot{q}) \in \mathbb{R}^{2\times 4}$ is the regression matrix; α, K stand for constant gain coefficients; $\hat{\Theta} = (\hat{\theta}_1, \ldots, \hat{\theta}_4)^T$, $\hat{\theta}_5$, $\hat{\theta}_6$ mean the best guess estimates of the unknown parameters Θ, θ_5 and θ_6. In order to exhibit the role of feed-back amplitude adjustable term $\frac{\kappa}{1-\rho}(\rho\|u_n\| + \mathcal{W})$ from (12) and (14) compared with constant gains $\hat{\Theta}$, $\hat{\theta}_5$, $\hat{\theta}_6$ of (15), estimations $\hat{\Theta}$, $\hat{\theta}_5$ and $\hat{\theta}_6$ are assumed in this experiment to be fully known, i.e., the following equalities are now fulfilled: $\hat{\Theta} = \Theta_n$, $\hat{\theta}_5 = \theta_{5,n}$ and $\hat{\theta}_6 = \theta_{6,n}$, respectively. In order to attain the convergence of task errors e less than or equal to 10^{-3} in approximately the same time, the following numerical values of gain coefficients for both controllers are taken: $K = 195$, $\alpha = 374.5$, $\lambda_0 = 1$, $\lambda_1 = 21$, $\lambda_2 = 16$, $\Lambda_n = 163$, $\kappa = 6$ and $\beta = \frac{5}{7}$, respectively. Moreover, mappings α_0 and α_1 may be estimated as $\alpha_0 = 2\|q - q(0)\| + 2\|y_1\|$, $\alpha_1 = 2\|y_1\| + 2\|y_2\|$. Observe that controller (15) requires the accurate measurement of $\dot{q}$. Under such condition, control law (15) from Ozbay et al. (2008) is shown to be asymptotically stable. The results of the simulations have shown that our controller generates tracking errors e, which are practically equal to zero for $T \geq 3.5$ whereas $e \neq 0$ for control law (15). Moreover, control scheme (14) provided smaller torques in the sense of Chebyshev's norm than that given by Eq. (15). Furthermore, at the beginning of the trajectory tracking for $t \in [0,\ 3]$, torques generated by control law (15) are clearly larger than those provided by our strategy (see Fig. 1). Secondly, both linearly and non-linearly parametrizable discontinuous friction terms of the form $5\text{sign}(\dot{q}) + 2\exp(-0.2\|\dot{q}\|^2)\text{sign}(\dot{q})$, exhibiting the Coulomb and Stribeck effects (Canudas de Wit et al., 1995), have been added to disturbed manipulator dynamic equations considered in the previous experiment. Moreover, the aim of this experiment is also to show robustness of control law (14) against singular movements. To force singular configurations, controller

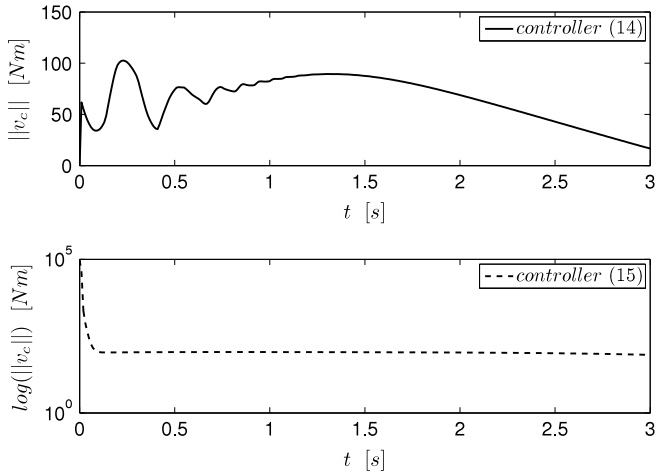

Fig. 1. Euclidean norm of v_c for controllers (14) and (15) with linearly parametrizable continuous friction terms—initial tracking phase.

(14) is to track the following trajectory: $p_d^2(t) = (0.2\cos(t),\ 0.68 + 0.2\sin(t))^T$. Let us note that p_d^2 generates infinitely many singular configurations corresponding to the set of time instances $\{t'_k :\ t'_k = \frac{\pi}{2} + 2\pi k,\ k \text{ is integer}\}$ (provided that $e = 0$) whose Lebesgue measure equals zero. The same controller (14) as that from the first experiment with the same controller gains has been applied in the second experiment. The results of the simulations are depicted in Figs. 2–4 which show stable finite-time convergence of the end-effector to desired trajectory p_d^2. Fig. 4 depicts the passing of the manipulator (coordinate q_2) through the singular manifold $\{q' :\ \det(J(q')) = 0\}$ at time instant $t' = \frac{5\pi}{2}$. Visible variations of e, v_c (Figs. 2–3) in the small neighbourhood of $t' = \frac{5\pi}{2}$ are a consequence of violating the assumption (4) in a set of time instances of non-zero Lebesgue measure by the corresponding numerical value of $\lambda_{\min}(J(q')M^{-1}(q')J^T(q'))$. Moreover, sudden increase of $\|v_c\|$ in a neighbourhood of $t = 5.1$ is a result of the Coulomb and Stribeck discontinuous friction terms. Although transient impetuous variations of $\|v_c\|$ (see Fig. 3) in a neighbourhood of both singular configurations and zero joint velocities still present absolutely continuous mappings, the control signal shown in Fig. 3 could not be feasible in the real case due to the physical limitations of the actuators. If this is the case, a smoothing of torques v_c in the aforementioned neighbourhoods should be carried out. In order to eliminate additional phase delay related with recursive low-pass filters which may further limit the performance of our closed-loop system (1), (13), (14), one could apply Newton predictor enhanced Kalman filter (NPEKF) (Han, He, & Xu, 2007) which provides a wide bandwidth and significantly reduces phase lag. In the third experiment, measured position q has additionally been contaminated by a measurement noise of a Brownian motion of the form $d\eta_i(t) = \begin{cases} 10^{-7}\sqrt{t}X(t)dt & \text{for } t \in [4,\ 6] \\ 10^{-6}\sqrt{t}X(t)dt & \text{for } t \in (6,\ 9] \end{cases}$; $X(t) \sim N(0,\ 1)$, $i = 1, 2$. In such case, we have tested our controller under conditions of the second experiment for trajectory p_d^1. The results of simulation are given in Fig. 5 which indicate a good performance of controller (13), (14) subject to measurement noise with two different normalized norm magnitudes. Two peaks of $\|\hat{e}\|$ in Fig. 5 are result of the Coulomb and Stribeck discontinuity friction terms.

5. Conclusions

A new class of task space absolutely continuous TSM controllers with the finite-time convergence property of the trajectory tracking by n-DoF rigid robotic manipulator has been proposed in this paper. Moreover, a novel TSM manifold, making it possible to simultaneously apply both the first and second order sliding mode control techniques with their advantages, was incorporated into control scheme. Although our dynamically computed torque technique needs knowledge about the system equations of the robot, the approach is able to handle uncertainty (in dynamics and disturbance) occurring in the system. It is worth to emphasize the fact that the absolutely continuous Jacobian transpose controllers proposed herein are able to cope with globally unbounded disturbances acting on the robotic manipulators as well as singular configurations which can potentially occur when tracking a desired trajectory by the end-effector.

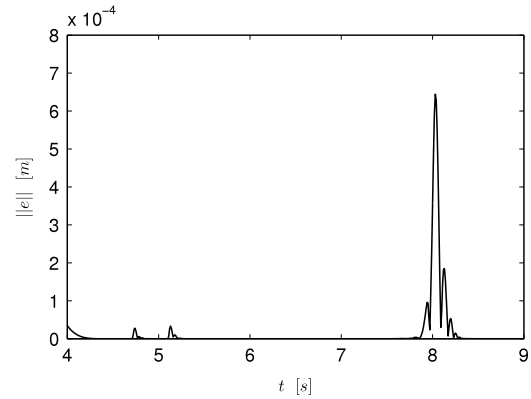

Fig. 2. Norm of task errors e for controller (14) in singular movement with both linearly and non-linearly parametrizable discontinuous friction terms.

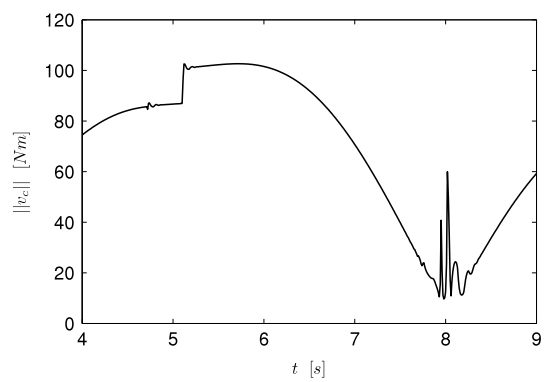

Fig. 3. Norm of v_c for controller (14) in singular movement with both linearly and non-linearly parametrizable discontinuous friction terms.

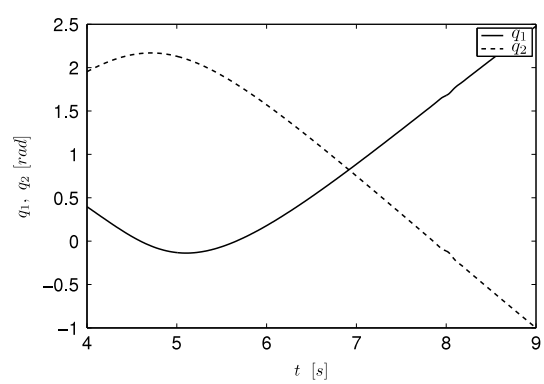

Fig. 4. Joint coordinates q for controller (14) in singular movement with both linearly and non-linearly parametrizable discontinuous friction terms.

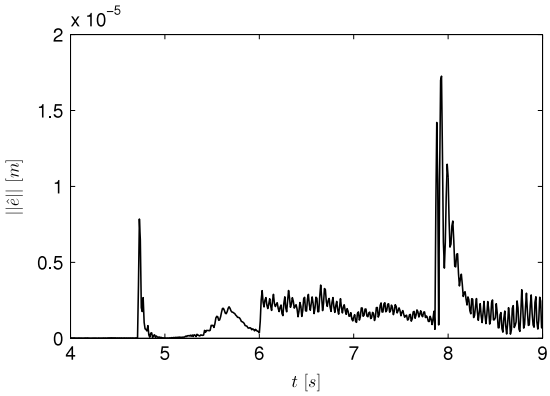

Fig. 5. Norm of estimated task errors $\|\hat{e}\|$ for controller (14) with measurement noise.

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